



Strength Training for Cross-Country Skiers

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Introduction

Remarkable changes have occurred in cross-country (XC) skiing in recent decades, with the introduction of races such as the sprint event and mass start, as well as better track preparation, improved skiing equipment, and changes in course profiles [1]. As an example, the race velocity in distance skiing (>10/15 km for women and men respectively) has increased by ~5–8% in World Cup races over the last decade and in sprint skiing (<1.8 km; ~3 min), a ~20% higher average velocity is evident compared to distance skiing [2]. Moreover, in World Championships and Olympics, five of six race events for each gender are sprints or mass starts and the placings are often determined by a final sprint. Therefore, the ability to conserve power for the final sprint and consequently perform supramaximal workloads is an important determinant of performance in today's XC skiing. Although aerobic endurance training has always been mandatory for skiers, the “high speed techniques,” such as double poling (DP) and the V2 skating technique, also demand well-developed upper-body power. Therefore, specific upper-body training, performed as speed, strength and muscular endurance training, has gained increased attention over the past decade [3–7]. This chapter therefore provides an updated synopsis of the demands for muscle strength in today's XC skiers, with a special focus on the upper body. It also provides an overview of the effect of training models in enhancing upper-body power, with a special emphasis on heavy strength training in endurance performance in highly trained XC skiers. In addition, alternative models such as sport-specific speed training and muscular endurance training are addressed. Finally, a practical application of efficient strength training is provided based on the distinctiveness of movement patterns in skiing techniques, and thus, the specificity of training for XC skiers.

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Performance in XC Skiing

XC skiing is a typical endurance sport where the average speed (m s^{-1}) for a required distance is mainly determined by energy turnover (J s^{-1}) and work economy (J m^{-1}) [8].

$$\text{Speed}(\text{m}\cdot\text{s}^{-1}) = \frac{\text{Energy turnover}(\text{J}\cdot\text{s}^{-1})}{\text{Work economy}(\text{J}\cdot\text{m}^{-1})}$$

Accordingly, a greater energy turnover (aerobic and/or anaerobic) and/or an improved work economy will increase speed and thus improve performance. Although it is widely accepted that aerobic energy turnover is closely related to performance in XC skiing (e.g., [9–11]), the importance of a high anaerobic energy turnover rate has gained increased attention in recent years (e.g., [12, 13]). In particular, sprints and mass starts require the ability to perform supramaximal workloads during parts of the races. Hence, the combination of a high aerobic power and anaerobic power/capacity coupled with an efficient technique is a distinctive feature of today's skiers [12, 13]. This complexity of energy sources used during races demands different training models to target specific adaptations. Although aerobic endurance training has always been mandatory, skiers have developed different strength training regimes over the years, aiming to optimize performance. It is well documented that skiers can initially increase their strength by 0.5–1.2% per session when they start systematic heavy strength training, despite their already large volume of endurance training [4, 6]. Further, increased cross-sectional area can be expected, at least in upper-body muscles, which seem as an important factor for achieving further strength gains [4]. However, in order to utilize the strength gain into enhanced skiing performance, the energy turnover and/or economy must be enhanced. The following section will discuss the potential effects of strength training on performance and show how strength training may influence skiers' energy turnover and work economy.

The Effect of Strength Training on Skiing Performance

An overview of the effect of heavy strength training on XC skiing performance is shown in Fig. 24.1. The early studies conducted by Hoff et al. [3, 14] and Østerås et al. [15] found large effects of heavy strength training on XC performance, while more recent investigations [4, 6, 16, 17] found mostly trivial effects. The reason for the discrepancy is unknown, but different outcome measures (i.e., time to exhaustion vs. constant duration tests) might explain some of the differences. In addition, testing equipment is an important factor when translating a specifically increased ability into actual performance. In most studies that found a “significant” effect of heavy strength training on performance (Fig. 24.1) [3, 4, 14, 15], a double poling ergometer was used. However, results from more sport-specific movements (such as rollerskiing outdoors) indicate that the effect of strength

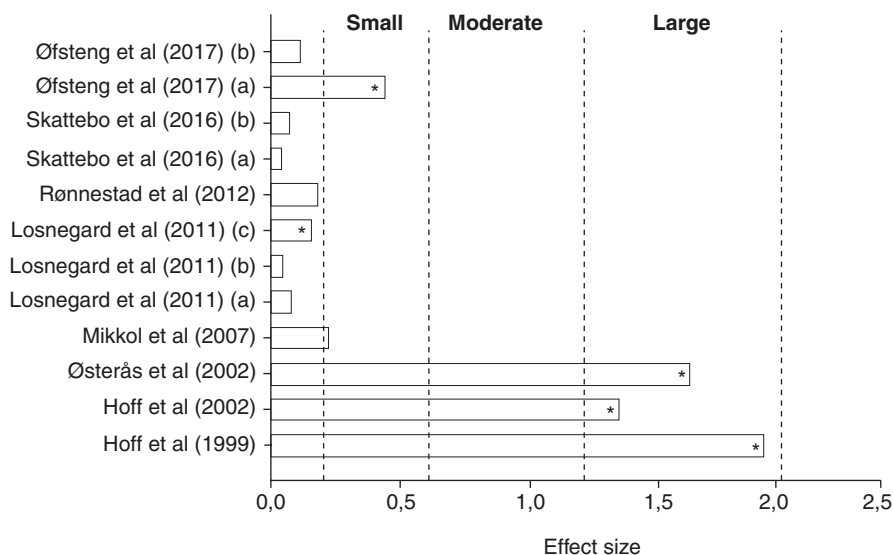


Fig. 24.1 The effect size of adding strength training to endurance training on performance in comparison to the control group. Calculated as the difference in change score between groups (raw data) standardized by the pooled standard deviation from baseline. *Significant differences from pre- to post-test between strength and control group ($P < 0.05$). Hoff et al. [3, 14] and Østerås et al. [15]: double poling ergometer with a time to exhaustion test (>4 min); Mikkola et al. [16]: 2-km time trial in double poling (>4.5 min); Losnegard et al. [4]: (a) 1.1 time trial classic style uphill (>4.5 min), (b) 1.3-km time trial freestyle uphill (>5 min), (c) double poling ergometer (5-min all-out); Rønnestad et al. [17]: 7.5-km time trial freestyle (>19 min); 6: (a) 3-min all-out double poling ergometer from “rested state”, (b) 3-min all-out double poling ergometer from fatigued state; [19]: (a) Time to exhaustion followed by a prolonged submaximal test in double poling on a roller ski treadmill. (b) Time to exhaustion in double poling in “rested state” on a roller ski treadmill

training is less pronounced [4, 16, 17]. This is probably attributed to the finding that rollerskiing outdoors demands a higher technical ability than a double poling ergometer, which in turn demands a higher technical ability than a strength test (e.g., more complex timing of force application). Thus, the effect of increased muscle strength on performance clearly reduces, and sometimes vanishes, when the exercise modes demand a higher degree of coordination of skiing movements.

The trivial effects shown in Fig. 24.1 could also be related to the duration of the performance tests. The nature of today’s competition format such as mass starts has increased the importance to conserve power for the final sprint, which has also been the interest in some recent studies [6, 18, 19]. Recently, Øfsteng et al. [19] investigated the effect of heavy strength training on DP performance in “rested state” and after a prolonged 90-min submaximal intensity. Although no significant effects were found in the “rested state”, the strength training group improved performance after submaximal intensity more than the control group (Fig. 24.1). Further, Børve

et al. [18] performed a similar study (DP performance test following 50-min sub-maximal test), but with focus on muscular endurance training (4×30 rep twice a week). Here, we found similar effects on DP performance as in Øfsteng et al. [19]. Taken together, these studies indicate that upper-body training (maximal strength or muscular endurance) could enhance “finishing abilities” after a prolonged exercise in XC skiers. However, the mechanisms behind these improvements are to date not clear and if these tests conducted on a treadmill are valid on snow needs to be further examined.

In a comprehensive study, aiming to investigate the effect of heavy strength training on XC skiing performance, we found a strong correlation between maximal strength and performance in females, whereas this relationship was trivial in men [4]. We argued that female skiers (and perhaps weaker male skiers) could benefit from adding strength training to their normal training. As differences in performance between gender are larger with exercises that involves a significant upper-body contribution [20], improved upper-body strength can potentially improve skiing performance more in female than in male skiers. Therefore, we conducted a new controlled study on elite junior female skiers (~16 years) and investigated their sprint abilities during both a rested 3-min all-out test and a 3-min all-out test after 27 min of DP on an ergometer [6]. Surprisingly, despite a significant increase in strength (~24% in 8 weeks), no beneficial but also no harmful effects of strength training was found on the 3-min performance tests, work economy, $\text{VO}_{2\text{max}}$, or 20-s all-out DP. We concluded that athletes and coaches should carefully consider whether young skiers should prioritize such training. Further, it was also found that skiers experienced the strength training as very demanding, indicating that an accumulated fatigue influenced the post-test. Thus, coaches should carefully plan and evaluate strength training in order to minimize the risk of unplanned “overload,” especially during the late preparation phase when the intensity of endurance training is increased. Hence, this study highlights what type of training young skiers should perform to optimize performance. In contrast to their senior counterparts, these athletes have limited time for training due to school and other activities in addition to the developmental factors accompanying puberty. It is well documented that aerobic energy turnover and work economy (including technique development) are the most important factors in achieving world-class level [9, 11, 13, 21, 22]. Clearly, these aspects must be prioritized and strength training may therefore be considered individually and perhaps differently (including endurance training) than in adult athletes.

For adult athletes, skiers have to plan their strength training in conjunction with their endurance training to maximize potential strength gains. Notably, strength training for XC skiers is preferably executed during the pre-season, while during the competition season body mass is reduced to optimize performance [22]. This is clearly challenging since muscle strength and muscle cross-sectional area are highly related [4]. Personal communications with world-class athletes and unpublished data from our lab indicate that the strength level is reduced in line with the anthropometrical changes.

Aerobic Energy Turnover

Aerobic energy turnover is the product of the maximal aerobic power ($\text{VO}_{2\text{max}}$) and the fractional utilization of $\text{VO}_{2\text{max}}$, also called the “performance VO_2 ”. A number of studies in various sports, including XC skiing, have shown that $\text{VO}_{2\text{max}}$ is not affected by heavy strength training e.g., [6, 17, 23]. The fractional utilization of $\text{VO}_{2\text{max}}$ is closely related to performance in numerous endurance sports such as running, on- and off-road cycling, and possibly XC skiing [24, 25]. Recent studies involving cycling suggested that the fractional utilization during a 40-min time trial was improved after adding heavy strength training [26]. Due to no changes in mitochondrial density, the authors proposed that the increased cross-sectional area of the muscles led to increased numbers of activated mitochondria and thus increased fractional utilization. In XC skiing, no data is available on the effect of strength training on fractional utilization $\text{VO}_{2\text{max}}$.

A unique aspect of XC skiing is the variety of techniques used during a competition, with different use of the upper and lower body. This affects the maximal aerobic power achieved in the sub-techniques, where skiers normally exhibit 3–10% lower peak aerobic power ($\text{VO}_{2\text{peak}}$) during ski skating and the DP technique than during running [27]. Therefore, for already highly trained XC skiers where the $\text{VO}_{2\text{max}}$ during a whole body exercise does not seem to change over time [22], reducing the “gap” between other techniques has been speculated to be one of the possible training benefits of increased upper-body training [28]. These aspects have received attention in recent decades, especially with increasing use of the V2 ski skating and DP techniques in certain terrains. The lower $\text{VO}_{2\text{peak}}$ in DP has been found to be related to intrinsic factors (e.g., O_2 -extraction) in the arm muscles compared to the legs, even in highly trained XC skiers [29]. It has also been proposed that the upper body has a great potential to increase its aerobic energy turnover rate, with a subsequently enhanced upper-body $\text{VO}_{2\text{peak}}$, in response to intensive endurance training [29]. However, to date no studies have been able to demonstrate such an effect [5, 18, 30]. In addition to intrinsic factors, the muscle mass involved is a limiting factor for $\text{VO}_{2\text{peak}}$ in many exercises [31], and in DP specifically [32]. Hegge et al. [32] investigated the $\text{VO}_{2\text{peak}}$ in DP with different muscle masses involved and found that the $\text{VO}_{2\text{peak}}$ increased sequentially with increased body mass contributions. Hence, theoretically, an increased muscle mass could increase the $\text{VO}_{2\text{peak}}$ and reduce the “gap” between sub-techniques such as DP or skating and running. Losnegard et al. [4] found that 12 weeks of heavy strength training increased trunk lean body mass in XC skiers. Interestingly, the $\text{VO}_{2\text{peak}}$ in skating also increased with no changes in running $\text{VO}_{2\text{max}}$. However, later studies have not verified this assumption [6, 17] and the findings should therefore be approached with caution. In summary, little data supports the suggestion that specific upper-body training in general (heavy, explosive or muscular endurance) leads to enhanced $\text{VO}_{2\text{peak}}$ in skiing sub-techniques.

Anaerobic Energy Turnover

Anaerobic energy turnover is divided into anaerobic power and anaerobic capacity. While anaerobic power is an important determinant of performance during short spurts of activity and acceleration, anaerobic capacity is more related to performance in middle distance competitions [33]. Due to the nature of XC skiing competitions, with repeated high-intensity periods, the O_2 -demand is well beyond VO_{2peak} during parts of the competition [12, 13, 34–37]. This influences the energy system contribution during a race and suggests that anaerobic power and capacity are important determinants of performance. Losnegard et al. [12] found a strong correlation between performance and accumulated oxygen deficit (ΣO_2 -deficit) in a simulated sprint race on a rollerski treadmill in a group consisting of long-distance (>50 km), distance (>15 km), and sprint skiers (<1.8 km). Moreover, sprint skiers have a higher body mass and body mass index than distance skiers due to the nature of the competition format and subsequently the race demands [2, 13]. Since ΣO_2 -deficit is related to the muscle mass involved in the exercise [38, 39], long-term heavy strength training with a subsequent increase in muscle cross-sectional area could increase the ΣO_2 -deficit. Moreover, the total volume of strength and speed training in international-level sprint skiers seems to be significantly higher compared to international-level distance skiers (~12% vs. 7% of total training time) in the preparation phase (6 months before the season), but similar with respect to total time of training during the year [2]. Thus, it can be suggested that some of the differences in ΣO_2 -deficit (and body mass) between skiers are related to differences in training focus regarding maximal strength, speed, and endurance training.

Several studies in XC skiing have investigated the relationship between “strength” and short performance tests where anaerobic power turnover represents the main energy source [6, 4, 13, 40–42]. Sandbakk et al. [13] investigated physiological and training differences between world-class and national class sprint skiers and found that acceleration (30-m test) and maximal strength did not differ between groups. This is supported by the findings of Losnegard et al. [4] who found no effect of heavy strength training on the acceleration phase during skating on roller skis. However, a tendency towards a reduced time during a 100-m test for the strength training group was found, indicating that strength training could potentially improve maximal speed [4]. Stöggl et al. [41] analyzed the relationship between general strength and “maximal speed” (test duration of ~60 s) with related kinetic and kinematical variables. They concluded that maximal strength had only a low to moderate correlation with “maximal speed,” whereas timing of the force application was the main discriminating factor between different levels of skiers. However, they proposed that strength ability is technique dependent, indicating that movement pattern is an important factor to consider during strength training.

Work Economy

The importance of work economy is emphasized by the finding of a close relationship with performance in groups of heterogeneous XC skiers (e.g., [43]). Further,

recent studies have shown that a within subject change in work economy (O_2 -cost), based on changes in equipment and technique, leads to enhanced speed at VO_{2max} and thus improved performance [44, 45]. Several studies in various sports indicate that both intrinsic factors (muscle fiber types, neuromuscular characteristics) and biomechanical factors (technique) contribute significantly to the variation in exercise economy seen between highly trained subjects (see review by [23]). Further, it has been shown that strength training influences work economy in endurance sports such as cycling [46, 47] and running [23]. In XC skiing, Mikkola et al. [16], Østerås et al. [15], and Rønnestad et al. [17] showed significant reductions in the O_2 -cost between pre- and post-tests in the strength groups, but the changes did not significantly differ from that of the control groups. Skattebo et al. [6], Carlsson et al. [48], Øfsteng et al. [19], and Losnegard et al. [4] found no effect of strength training on work economy on a double poling ergometer, double poling or ski skating on a rollerski treadmill. In addition, Hoff et al. [3, 14] used unconventional methods to measure work economy and it has recently been argued that they actually did not measure the O_2 -cost (see Skattebo et al. [6] for details). Overall, no study has found a convincing beneficial effect of heavy strength training on work economy in XC skiing. The discriminating findings between endurance sports might be related to the fact that XC is an extremely technically demanding exercise, involving both arms and legs to different extents. Moreover, in elite skiers it has been shown that the O_2 -cost is typically reduced by ~3% during the preparation phase with a “traditional Norwegian training regime.” In this period, skiers included ~7% strength training as part of the total training volume [22], although it should be noted that no clear evidence for a “cause-effect” exists. Nevertheless, the reduced O_2 -cost corresponds to 0.5% per month or 1–1.5% during a typical strength training intervention period (8–12 weeks) used in research. This implies that an eventual effect of strength training is difficult to detect, and that future research should use appropriate methods to evaluate whether such effects exist.

Practical Recommendations

In order to elicit the desired physical adaptations, it is important to evaluate correct muscle use and movement pattern before implementing strength training. In XC skiing, DP and V2 skating are the most-used techniques in races and are both very technically demanding exercises. Here, the joints are engaged in a sequential pattern before and during the poling phase to optimize propulsion and transfer potential and rotational energy as forward kinetic energy (e.g., [44, 49]). The upper-body muscles are therefore engaged in a “first in-first out pattern” involving (I) the abdominal muscles and hip extensors, followed by (II) the shoulder extensors latissimus dorsi, teres major, delta and pectoralis major, and (III) the shoulder and elbow extensor triceps brachii [49–51]. Such timing of force application is, thus, considered one of the most important characteristics of effective technique and resulting performance [41, 45]. To optimize strength training it is therefore necessary to (I) elicit an exercise mode covering the correct muscle use and (II) simulate the specific movement

patterns used in the respective techniques. During indoor maximal strength training, Losnegard et al. [4] used two relatively simple exercises: “seated pull-down” and “standing double poling.” These two exercises served as main exercises and supplemental exercises for core stability (including the use of Swiss balls, sling exercises, and various mat exercises) as additional training models either before or after the main program.

As stated previously, it is generally agreed that “stronger is not necessarily better” in optimizing performance in XC skiing [4, 6, 13, 41]. Hence, for skiers who already have high strength levels, the overall goal might be to maintain maximal strength and improve other physiological or technical mechanisms. One factor of interest is the metabolic response during DP, which seems to be different for the arms and legs [29, 52, 53], indicating that intense upper-body endurance training may increase the arms’ ability to extract oxygen and thus enhance DP performance. To enhance muscular endurance, the number of repetitions is high, while the loads are relatively low (20–100 reps/set; [54]). Such training for endurance performance has received attention in various sports including running [55, 56] and rowing [57, 58] as it seems to target different muscular and neurological adaptations compared to maximal strength training [54]. Specifically for XC skiing, Nilsson et al. [5] showed that 20-s or 180-s interval training in a DP ergometer increased both 30-s and 6-min power output in well-trained XC skiers, while Vandbakk et al. [30] demonstrated increased time to exhaustion after 8 weeks of 30-s DP intervals. The short duration speed training (“explosive”; >30 s) is a widely used training model for competitive XC skiers and could serve as a training model to maintain or enhance performance. However, such training methods are relative new in XC skiing, and further studies are encouraged to develop such models (combined with technique training) directly for XC skiing.

Carlsson et al. [48] tested two different training regimes in junior skiers (adding heavy strength training or interval training on a ski ergometer to their normal training routines) and found that both groups improved maximal speed during DP on a treadmill. However, there were no differences between groups in DP $\dot{V}O_{2peak}$, maximal speed, or work economy. Moreover, Børve et al. [18] showed that replacing part of high-intensity running intervals with training using the “standing double poling” exercise, 4×30 reps twice a week, improved well-trained skiers’ finishing abilities during a simulated race on a rollerski treadmill. Further, even though the aim of the training was to increase muscular endurance, the strength gain (1RM) per session was in the range to studies using heavy strength training for XC skiers with similar strength exercise mode (0.6% vs. 0.5–1.2% per session) [4, 6, 17]. This implies that low-resistance strength training with high mobilization could be an alternative to heavy strength training in well-trained skiers, at least for short-term adaptations prior to competitions or during short block periodization. Together, these studies indicate that short-term upper-body endurance training may be a promising training model that could have direct applications for well-trained skiers aiming to improve their DP performance. From a practical point of view, outdoor rollerskiing with extra weight (e.g., a weight-west), uphill

Table 24.1 Example of a 6-week training program for heavy strength training

Week	1–2	3–4	5–6
Day 1 (sets/reps)	4/10RM	4/8RM	4/6RM
Day 2 (sets/reps)	4/6RM	4/5RM	4/4RM

Note: RM; repetition maximum. When athletes manage more repetitions than planned, the loads increase. Test of 1RM is performed before and after the training period

Table 24.2 Example of a 6-week muscular endurance training program

Week	1–2	3–4	5–6
Day 1 (sets/reps)	4/40	4/35	4/30
Day 2 (sets/reps)	4/30	4/25	4/20

Note: Loads represent 60–70% of 1RM, depending on the level of the athletes. Tests of 1RM/60% of 1RM are performed before and after the training period

skiing or slow resistance (manipulation of rollerski wheels) might serve as appropriate exercise modes to enhance sport-specific “muscular endurance training” for XC skiers.

A detailed training program for strength and muscular endurance training, respectively, as used by world-class skiers, is presented in Tables 24.1 and 24.2. Both programs consist of a 6-week period in a linear periodized fashion.

Summary

This chapter provides an updated review, based on scientific evidence, of the specific needs and effects of strength training on performance in high-level cross-country skiing. Based on the available literature, a certain level of strength is necessary for optimizing performance in XC skiing, based on skiers’ race preferences and techniques. In sprint skiing, a higher level of strength seems necessary than in distance skiing due to the ~20% higher average speeds and very high intermittent workloads. Further, the strength level seems to be dependent on technique with respect to different contributions from the upper and lower body. Specifically, the V2 ski skating and classic DP techniques are the major techniques used in today’s races. These techniques demand well-developed upper-body power; therefore, developing specific upper-body training seems important. However, most studies investigating the effect of heavy strength training have shown trivial effects on performance in XC skiing, implying that a substantial individual variation exists. These findings emphasize that in general “stronger is not necessarily better” in optimizing performance in XC skiing. Coaches and athletes are therefore encouraged to develop training programs that include: (I) ways to combine strength training with skiers’ endurance training to minimize the risk of unplanned overload and subsequently achieve optimal performance development; (II) training stimuli (heavy, explosive or muscular endurance) based on the needs of the individual skier; (III) sport-specific strength exercises designed to stimulate correct muscle use in the required

techniques, but simple enough to achieve the necessary overload; and (IV) ways to translate the movement patterns and timing of force applications needed in specific techniques to strength exercises to optimize technique.

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